E06 Post-Processing

CRE132 worksheet

What you'll make

A dark arena lit by burning braziers, where the brightest parts glow softly (**Bloom**) and the screen darkens towards the corners (**Vignette**). Then a small script that switches each effect off and on with the number keys, so you can see what it adds.

Post-processing changes the whole finished picture, like a filter on a photo. By the end you'll know how URP's **Volumes** and **Volume Profiles** work, why bloom needs something "brighter than white", and how a script can find and change an effect.



Figure 1: The finished scene: the arena at night, lit by four braziers, with a soft glow and dark corners.

Before you start

1. Unzip `CRE132_E06_PostProcessing_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_E06_PostProcessing_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Effects/E06_PostProcessing`.
3. Press **Play** and walk about (WASD or the arrows). The arena is dark and the braziers light the ground round them, but the picture is flat: nothing glows and the corners are as bright as the middle. You'll add both effects, then a switch for them. Press Play again to stop.

The scene has a **Global Light 2D** turned down to **Intensity** 0.3 (night), and four **Braziers**, each with a child **Light** (a 2D point light, **Intensity** 1.8). There's no post-processing yet.

Making a script: in the Project window, go to the folder, right-click and choose **Create > Scripting > MonoBehaviour Script**. Type the name straight away and press Enter. The name must match the class name in the code exactly, capital letters included.

1. A Volume and a profile

Post-processing comes from a **Volume**, which holds a **Volume Profile**. A profile holds the effects, called **overrides**.

1. Choose **GameObject > Volume > Global Volume** from the menu bar. A **Global Volume** appears in the Hierarchy. Its **Volume** component has **Mode: Global**, so it affects the whole screen, but its **Profile** slot is empty.
2. In the Project window, open `Assets/_game`. If there's no `Settings` folder in it, right-click `_game`, choose **Create > Folder** and name it `Settings`. Right-click the `Settings` folder and choose **Create > Rendering > Volume Profile**. Name it `E06 Post Processing`.
3. Select the **Global Volume** again and drag `E06 Post Processing` from the Project window into its **Profile** slot. The Volume now shows an **Add Override** button.
4. Click **Add Override > Post-processing > Bloom**. A **Bloom** section appears. Tick the box to the left of each setting you want to change, then set it (see the table).
5. Click **Add Override > Post-processing > Vignette** and set it the same way.

Override	Setting	Value
Bloom	**Threshold**	0.9
	Intensity	1.5
	Scatter	0.7
Vignette	**Intensity**	0.4
	Smoothness	0.4

- **Bloom** makes bright parts glow and spill light onto their surroundings. **Threshold** 0.9 means only things brighter than 0.9 glow; **Intensity** is how strong the glow is; **Scatter** is how far it spreads.
- **Bloom needs something bright.** Colours normally stop at 1 (white), but the project uses **HDR** (in the URP settings), so values can go higher. The brazier lights have **Intensity** 1.8, so the ground they light is "brighter than white", and that's what blooms. The dark arena (0.3) doesn't.
- **Vignette** darkens the edges of the screen, which pulls the eye towards the middle. **Smoothness** is how soft the edge of the darkening is.

Test it: press Play. Still no glow and no dark corners. The Volume is ready, but the camera isn't using it yet.

2. Switch post-processing on for the camera

Select the **Main Camera**. In its **Camera** component, open **Rendering** and tick **Post Processing**.

Test it: press Play. The ground round each brazier glows softly, and the corners of the screen are darker. Untick **Post Processing** while playing and both effects disappear; tick it again. Try **Bloom Intensity** 5 to see how quickly strong effects turn muddy, then put it back to 1.5.

- **The camera's Post Processing box is off by default** in a new scene. It's the usual reason "post-processing doesn't work".
- **A Volume on its own does nothing to a camera** that doesn't have Post Processing ticked, and a ticked camera with no Volume has nothing to show.

3. Switching effects: PostProcessToggle

Make a folder `Assets/_game/Scripts/E06_PostProcessing` and create `PostProcessToggle` in it:

```
// Switches the Volume's effects on and off with the number keys, to compare with and
// without: 1 = Bloom, 2 = Vignette. Each effect is an "override" in the Volume Profile
// with an active switch.
using UnityEngine;
using UnityEngine.InputSystem;
using UnityEngine.Rendering;
using UnityEngine.Rendering.Universal;

public class PostProcessToggle : MonoBehaviour
{
    [SerializeField] private Volume volume;

    private Bloom bloom;
    private Vignette vignette;

    private void Start()
    {
        // TryGet finds an effect in the profile; it's false if the profile doesn't have
        // one.
        volume.profile.TryGet(out bloom);
        volume.profile.TryGet(out vignette);
    }

    private void Update()
    {
        // Reading keys directly from the Input System, which is fine for a quick test
        // tool.
        if (Keyboard.current.digit1Key.wasPressedThisFrame && bloom != null)
        {
            bloom.active = !bloom.active;
            Debug.Log("Bloom " + bloom.active);
        }
        if (Keyboard.current.digit2Key.wasPressedThisFrame && vignette != null)
        {
            vignette.active = !vignette.active;
            Debug.Log("Vignette " + vignette.active);
        }
    }
}
```

- **Two new `using` lines:** `UnityEngine.Rendering` has `Volume`, and `UnityEngine.Rendering.Universal` has URP's effects, `Bloom` and `Vignette`.
- **`TryGet(out bloom)`** looks in the profile for a Bloom override and puts it in `bloom`. If the profile has none, `bloom` stays `null`, which is why `Update` checks `bloom != null`.
- **`bloom.active = !bloom.active`** flips the effect's on/off switch: `!` means "not", so true becomes false and false becomes true.
- **`Keyboard.current.digit1Key`** reads the **1** key directly. That's fine for a quick test tool; for real game controls, use an action in the Input Actions asset.

Add it to the **Global Volume** and drag the **Global Volume** itself into the **Volume** slot.

Test it: press Play, then press **1** (the number keys along the top, not the number pad) to switch Bloom off and on, and **2** for the Vignette. The Console says which is on.

Check your work

- ☐ The arena is dark, and the ground round each brazier glows softly.
- ☐ The corners of the screen are darker than the middle.
- ☐ **1** switches the glow off and on; **2** switches the dark corners off and on.
- ☐ The **Global Volume** uses your `E06 Post Processing` profile, with **Bloom** and **Vignette**.
- ☐ **Post Processing** is ticked on the Main Camera; unticking it removes both effects.
- ☐ Compare with `CRE132_E06_PostProcessing_Finished` if something behaves differently.

Key points

- **Post-processing is cheap mood.** A dim scene plus bloom and a vignette reads as "night" or "dungeon" straight away.
- **Three things must be in place:** a **Volume** with a **Profile**, overrides in the profile, and **Post Processing** ticked on the camera.
- **Bloom needs something bright** to work with: lights above 1, or very bright sprites.
- **A little goes a long way.** Strong effects quickly look muddy.

If something goes wrong

- **No effects at all:** the Main Camera's **Rendering > Post Processing** is unticked, or the Volume's **Profile** slot is empty.
- **There's no Add Override button:** the Volume has no **Profile** yet. Drag `E06 Post Processing` into the slot.
- **The effects show only near the middle, or not at all, with a profile in place:** the Volume's **Mode** is **Local**. Set it to **Global**.
- **An override does nothing:** the tick box next to the setting isn't ticked, so the profile isn't changing it.
- **No glow anywhere: Bloom Intensity** is still 0, or **Threshold** is higher than anything in the scene.
- **Pressing 1 does nothing:** the Game view needs to be focused (click it), or the profile has no Bloom (the Console shows nothing).
- **UnassignedReferenceException in PostProcessToggle:** the **Volume** slot is empty.
- **"The type or namespace name 'Bloom' could not be found":** the `using UnityEngine.Rendering.Universal;` line is missing.

Try it

1. Add **Color Adjustments** to your profile (**Add Override > Post-processing > Color Adjustments**): lower **Saturation** and give it a blue **Color Filter** for a cold night.
2. Make a local Volume: set a second Volume's **Mode** to **Local**, add a **Box Collider** (the 3D one: Volumes don't use 2D colliders) with **Size Z** 30 so it reaches the camera, and give it a red-tinted profile. The tint shows only when the camera is inside the box, like a lava room.
3. Add a third key to PostProcessToggle for your Color Adjustments.
4. Harder: pulse the vignette red when the player's health is low (T08). You'll need `vignette.color.value` and `vignette.intensity.value`.